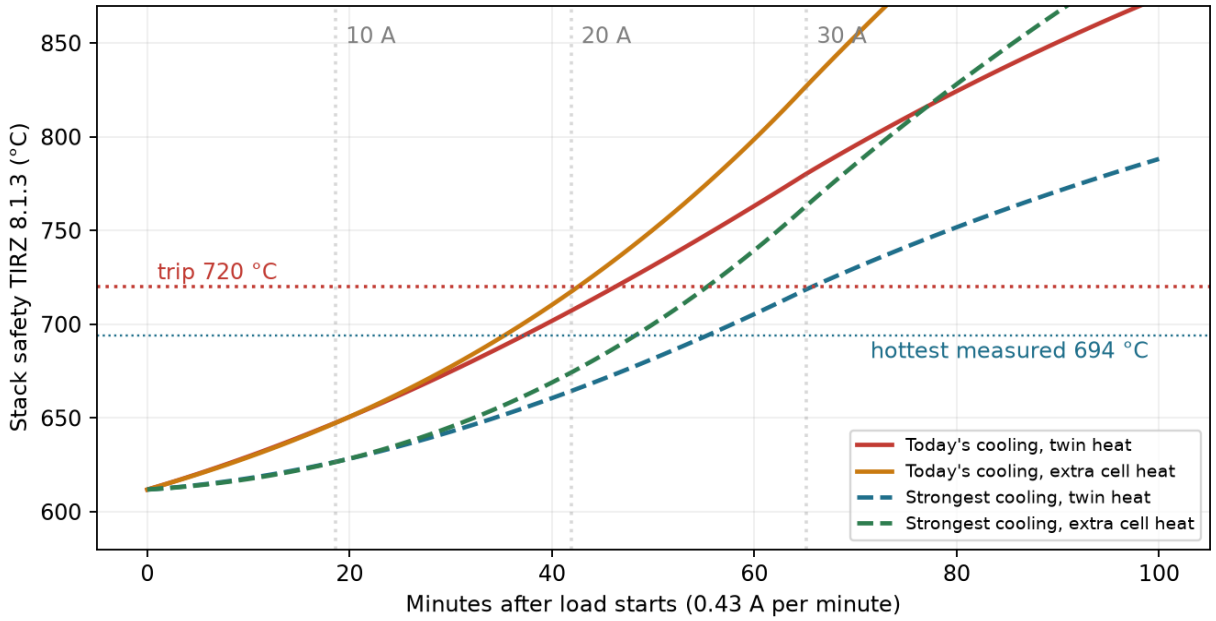
Rated power temperatures

SOFC POC plant: what happens to the stack and heater temperatures at 30 A per string (60 kW), and which trips break. 2026-11-05. Same content as the online pages.

Contents: 1 The answer · 2 How the stretch works · 3 Every trip sensor

1 At 60 kW, would the plant trip?

Yes. In every case the stack safety thermometer passes its 720 °C trip on the way up, at about 20–30 A per string (40–52 kW), 43–66 minutes after load starts. The exhaust trip (200 °C) comes next. The heaters stay well below their own trip values.



Solid = air settings used in part load (9,000 L/min, air-in 600 °C); dashed = strongest cooling the twin allows (10,000 L/min, 500 °C). Two ways of counting cell heat (page 2).

Case	Stack trip after	Current	Power	Exhaust trip after	Would settle at
Today's cooling, twin heat	47 min	22 A	43 kW	60 min	1042 °C
Today's cooling, extra cell heat	43 min	20 A	40 kW	52 min	1535 °C
Strongest cooling, twin heat	66 min	30 A	52 kW	87 min	880 °C
Strongest cooling, extra cell heat	56 min	26 A	47 kW	66 min	1186 °C

There is no safe resting point at 30 A with the present air flow: the stack would keep heating to 880–1,500 °C. Holding it under the trip needs roughly 13,000–23,000 L/min of air, 1.3–2.3 times the plant's maximum.

If you remember one thing: with the cooling the plant has, 30 A trips the stack's 720 °C safety at around 40–52 kW.

2 How the stretch from 16 A to 30 A works

> We treat the stack as a tank of heat. Each minute we add the heat the cells and the gas put in, take away what the air carries out, and see where the temperature goes. Every other thermometer is then read off the stack temperature and the current.

Step 1 Voltage line (earlier pages): each string loses about 5 V per extra amp, so about 345 V at 30 A and 51 kW. **Step 2 Fuel rule**: gas follows current by chemistry (Faraday), about 170 L/min. **Step 3 Heat tank** (new): heat in minus heat out gives the stack °C each minute. **Step 4 Sensor ties** (new): each thermometer as a straight line of stack °C and current.

At 690 °C and 30 A	°C per hour
Cells making current	+198 to +350
Natural gas 170 L/min	+63
Air heater share	+8
Cathode air carries out	–52 to –129
Lost through walls	–8
Net change	+130 to +360

The cell heat at 30 A is the one number the data cannot pin down. **Twin heat**: grows in step with current; replays the 10 real part-load runs to 2.7 °C. **Extra cell heat**: grows faster, because each amp heats more as the voltage drops (physics); replays to 2.9 °C. Both fit the measured runs, which never passed 16 A, so both are shown as a range.

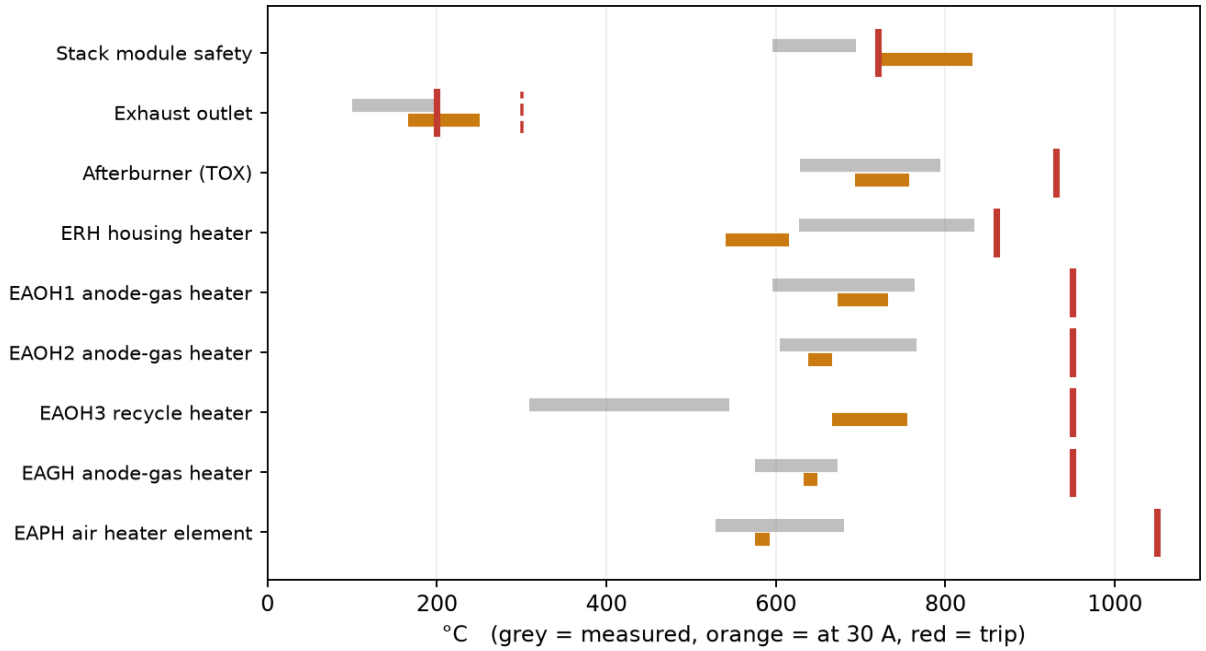
Each heater and safety line was tested by leaving one real run out and predicting it.

Scripts: session scratchpad folder rated/ (load.py, rated.py, replay.py, make.py); twin numbers from 05_twin_2d/app.js.

If you remember one thing: at 30 A more heat goes in than the air can take out, in both versions, so the temperature climbs.

3 Every trip sensor at 30 A

Two trips break: stack safety (720 °C) in all four cases and exhaust (200 °C) in three of four. The seven heater and afterburner safeties stay at least 170 °C below their trip values.



Sensor	Hottest measured	At 30 A	Trip	Left-out run miss
Stack module safety (TIRZ 8 1 3)	694 °C	721–832 °C	720 °C	±5 °C
Exhaust outlet (TIRZ 8 1 10)	204 °C	166–250 °C	200 °C	±24 °C
Afterburner (TOX) (TIRZ 8 1 9)	794 °C	693–758 °C	930 °C	±36 °C
ERH housing heater (TIRZ 8 2 5)	834 °C	540–616 °C	860 °C	±86 °C
EAOH1 anode-gas heater (TIRZ 8 2 2)	763 °C	673–732 °C	950 °C	±34 °C
EAOH2 anode-gas heater (TIRZ 8 2 3)	766 °C	638–666 °C	950 °C	±30 °C
EAOH3 recycle heater (TIRZ 8 2 4)	545 °C	666–755 °C	950 °C	±29 °C
EAGH anode-gas heater (TIRZ 8 2 1)	673 °C	633–649 °C	950 °C	±19 °C
EAPH air heater element (TIRZ 8 1 1)	680 °C	576–593 °C	1050 °C	±21 °C

Heaters are held at their own set temperatures and turn down when the stack makes its own heat, so they hardly follow the load. The ERH housing heater already reached 834 °C (trip 860) in part load on its own setting. The exhaust trip was raised to 300 °C on 27 March 2026.

Air needed to hold the stack safety at 700 °C at 30 A: air-in 600 °C 31,700–42,900 L/min; 500 °C 17,400–23,500; 400 °C 12,700–17,200. Plant maximum about 10,000 L/min. (Air rule fitted only up to 10,000.)

Not covered: the plant's real controller at 30 A, and the 0.72 V cell-voltage limit that already stops current near 15 A. Next step: ask the stack maker for the 60 kW air flow and air-in temperature; run a short 20 A test step with air at maximum.

If you remember one thing: only the stack and exhaust trips are reached; rated power needs 1.3–2.3 times the air the plant can supply, or cooler air-in.